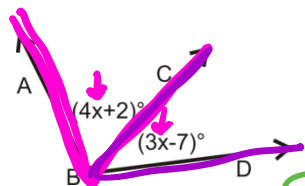


More of Lesson 1-5 - Angle Bisector

Warm - Up: First, find the value of x if $m\angle ABD = 107^\circ$. Next, find the measure of each angle.



$$4x + 2 + 3x - 7 = 107$$

$$7x - 5 = 107$$

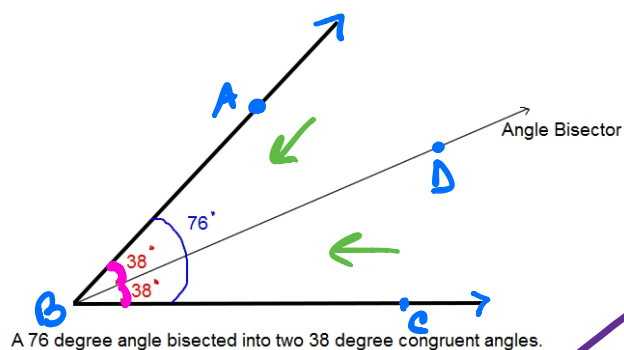
$$x = 16$$

$$m\angle ABC = 66^\circ$$

$$m\angle CBD = 41^\circ$$

Angle Bisector

<http://www.mathopenref.com/constbisectangle.html>



$\vec{BD}$ bisects $\angle ABC$

$$m\angle ABD = m\angle DBC$$

(numbers in degrees)

$$\angle ABD \cong \angle DBC$$

(shapes are same size)

Congruent

Example 1:

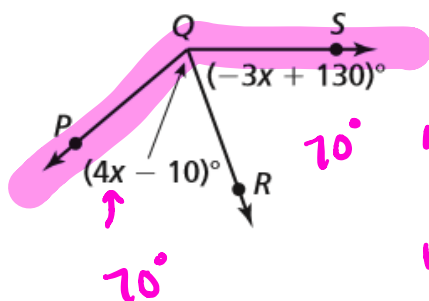
$\vec{QR}$ bisects $\angle PQS$.

Find $m\angle PQR$ and $m\angle PQS$.

Set = to each other

$$4x - 10 = -3x + 130$$

$$x = 20$$

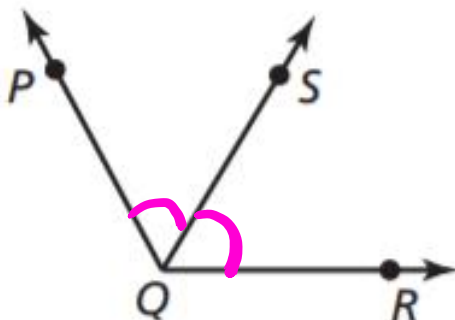


$$70^\circ \quad m\angle PQR = 70^\circ$$

$$m\angle PQS = 140^\circ$$

Example 2: This is found on page 45 in your book.

In Exercises 33–36, $\overrightarrow{QS}$ bisects $\angle PQR$. Use the diagram and the given angle measure to find the indicated angle measures. (See Example 5.)



33. $m\angle PQS = 63^\circ$. Find $m\angle RQS = 63^\circ$ and $m\angle PQR = 126^\circ$.

34. $m\angle RQS = 71^\circ$. Find $m\angle PQS = 71^\circ$ and $m\angle PQR = 142^\circ$.

35. $m\angle PQR = 124^\circ$. Find $m\angle PQS$ and $m\angle RQS$.

36. $m\angle PQR = 119^\circ$. Find $m\angle PQS = 59.5^\circ$ and $m\angle RQS = 59.5^\circ$.